

Insurance Price Optimization & Smoothing Spiky Demand Models

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- **Share how Actuaries Build & Optimize Policy Insurance Premium**
- **Share a technique to Smooth Spiky Models, turning the predicted Demand Curves into differentiable curves.**
- **Brainstorm the parallel to AUTO1 and which learnings can be transposed and applied**

- 01 **How is a Car Insurance Premium Calculated?**
- 02 **How are Prices Optimized According to Demand?**
- 03 **How to Smooth Models for Optimization?**

How is a Car Insurance Premium Calculated?

How do Actuaries do it?

Premium needs to take into account:

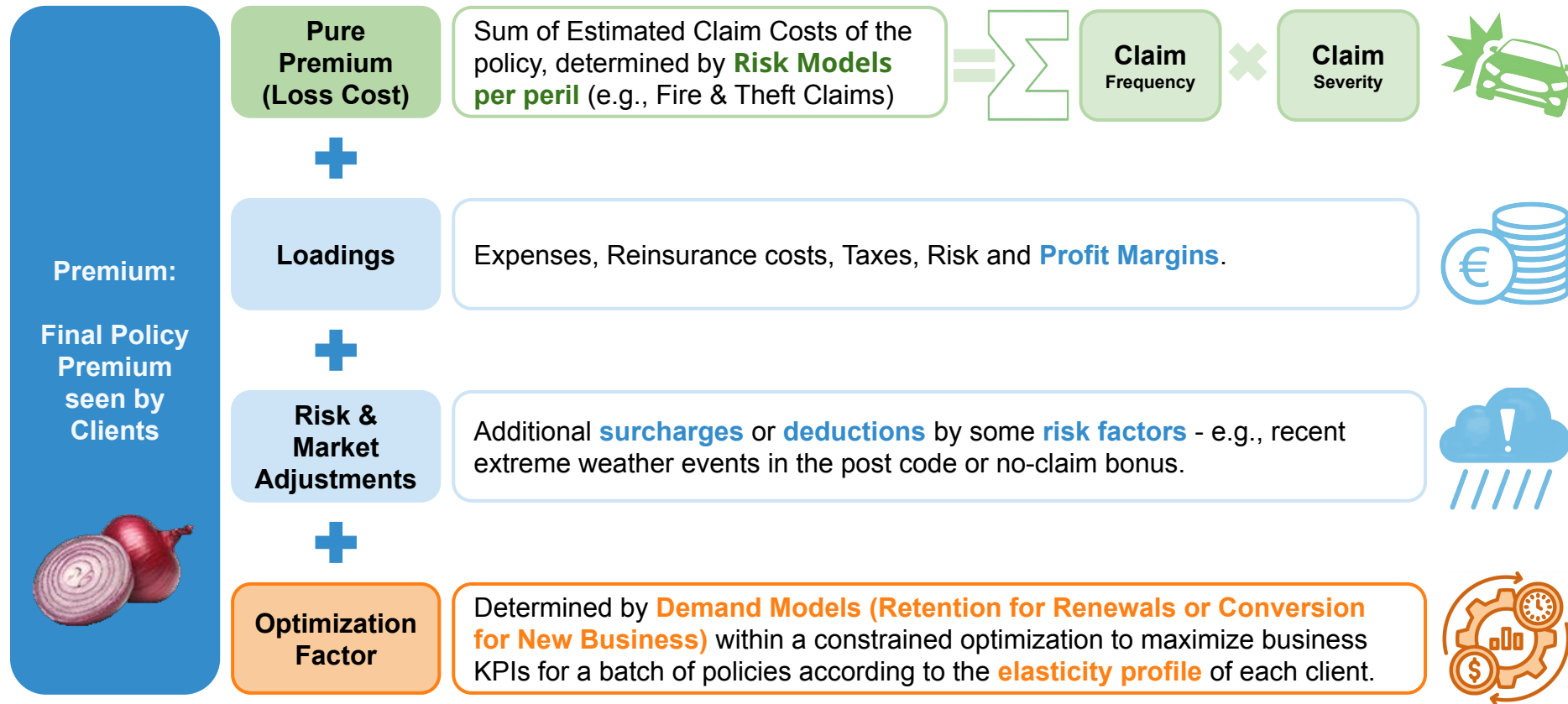
- How likely is the customer to have an **accident (claim frequency)**?
- If there is an **accident**, how much will it **cost (claim severity)**?
- What **other expenses or adjustments** need to be accounted for (**loadings**)?
- If we increase the price, how likely is the customer to **churn**? What is the expected **Lifetime Value (LTV)** of the customer? (**Demand**)
- Looking at the entire portfolio, how much are the total expected costs (**portfolio optimization**)?

A basic approach consists in **Pricing to Loss Cost**, i.e., predicting the estimated claim cost of the policy and adding additional loadings to cover other costs and a profit margin. But it lacks **sensitivity to Customer** (and **Agent**) **Behaviour** and their Price Elasticity.

The challenge is to automatically adapt the price based on **Risk Profile, Client's Elasticity and Business KPIs**.



Premium Components



How are Prices Optimized According to Demand?

Insurance Price Optimization Problem Definition

$$\text{Premium} = \text{Pure Premium} + \text{Loadings} + \text{Risk \& Market Adjustments} + \text{Optimization Factor}$$

To simplify, let's exemplify with **Retention Models** for Renewals and call:

- **Premium** to the final policy price the client sees;
- **Costs** to all other components except for Profit Margin.

The Insurance Price Optimization Problem consists in determining the set of policy **Premiums** that **maximize**

$$\text{Margin} = \sum_{\text{Policies}} (\text{Premium} - \text{Costs}) \times \text{Retention Probability}$$

subject to:

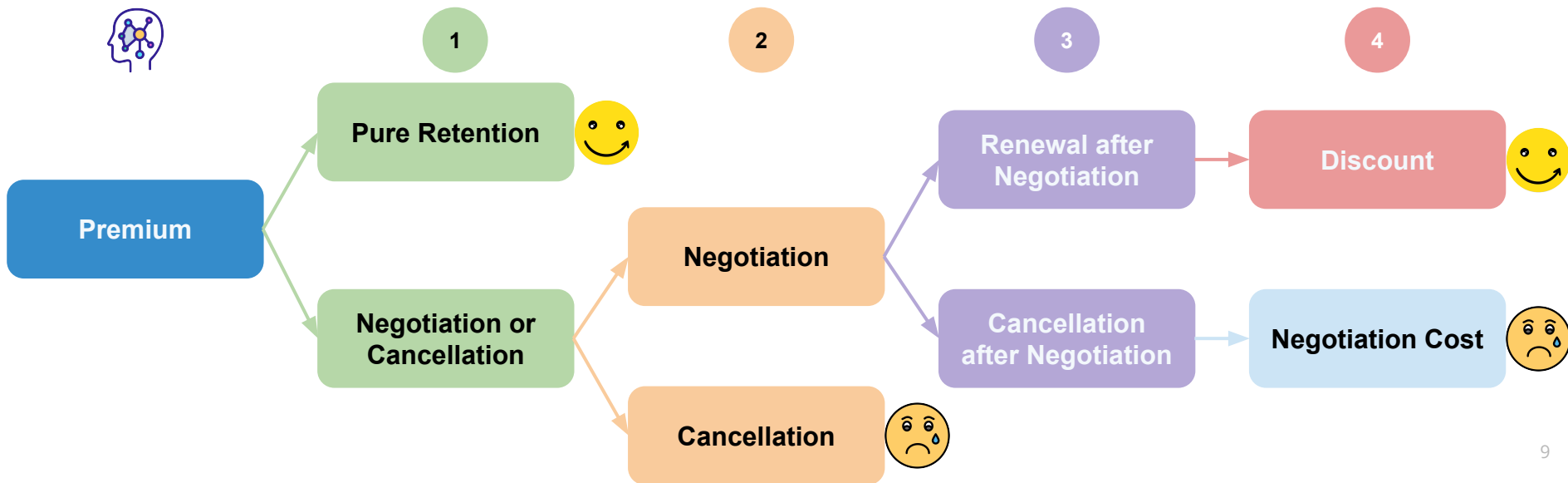
- **Global Constraints:** overall rate change, e.g., average increase of 10%, i.e., the sum of optimized premiums will be the sum of the expiring premiums plus 10%.
- **Local Constraints:** Caps & Collars to determine the maximum and minimum prices or rate changes per segment, i.e., to define the Optimization Space per policy. For instance, **clients without claims** with a **good loss ratio** have much **smaller increases** than clients with recent claims and high loss ratio.

The **Retention Probability** is given by **Demand Models** and it changes according to the **Premium**.

More Complex Renewal Price Optimization Design

In case of **Negotiation**, this is an example of an Optimization Design, where Pure Retention denotes a renewal without negotiation request. The example below uses **3 Classification** (1-3) and **1 Regression Model** (4) for the Demand Component. Costs include other models which depend on the policy characteristics (driver, car, coverages,...), not on the final price itself, so they are "static" for optimization purposes.

$$\begin{aligned} \text{Margin} = & \sum (\text{Premium} - \text{Costs}) \times \text{Pure Retention Probability} \\ & + \sum (\text{Premium} - \text{Costs} - \text{Discount}) \times (1 - \text{Pure Retention Probability}) \times \text{Negotiation Probability} \times \text{Renewal After Negotiation Probability} \\ & - \sum \text{Negotiation Costs} \times (1 - \text{Pure Retention Probability}) \times \text{Negotiation Probability} \times (1 - \text{Renewal After Negotiation Probability}) \end{aligned}$$



Optimization Constraints

Example of Global Constraint: $\sum \text{Optimized Premium} = 1.1 \times \sum \text{Expiring Premium}$

Example of Local Constraints: Caps & Collars by Package and Expiring Loss Ratio (ELR), defined by the tables below:

Collars	Package		
ELR	TPLFT	Comprehensive with Deductible	Comprehensive without Deductible
[0%, 10%)	-4%	-2%	-1%
[10%, 20%)	-2%	-1%	0%
[20%, 30%)	0%	0%	1%
...	...	...	...
[90%, 100%)	10%	10%	8%
[100%, 110%)	12%	12%	10%
...	...	...	...
[200%, +∞)	20%	20%	20%

Caps	Package		
ELR	TPLFT	Comprehensive with Deductible	Comprehensive without Deductible
[0%, 10%)	4%	5%	5%
[10%, 20%)	6%	7%	8%
[20%, 30%)	8%	9%	10%
...	...	...	...
[90%, 100%)	36%	40%	38%
[100%, 110%)	38%	42%	38%
...	...	...	...
[200%, +∞)	48%	46%	38%

Final Result Evaluation

After the optimization is ready, all other business KPIs must be recomputed for the final **Optimized Premium**, namely:

- **Profitability**, given by

- **Expected Loss Ratio:**

$$\frac{\sum_{\text{Policies}} \text{Loss Cost}}{\sum_{\text{Policies}} \text{Premium Retention}}$$

- **Growth**, given by

- **Expected Policy Retention:**

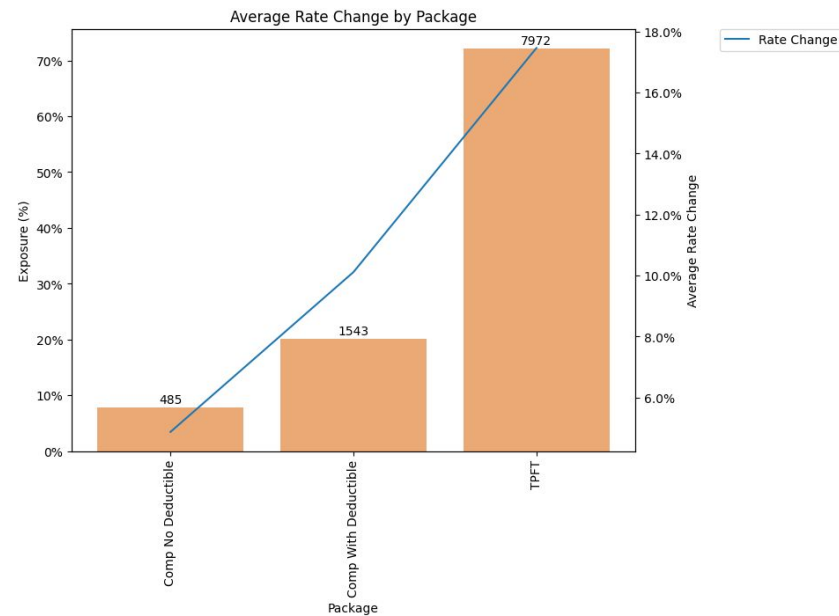
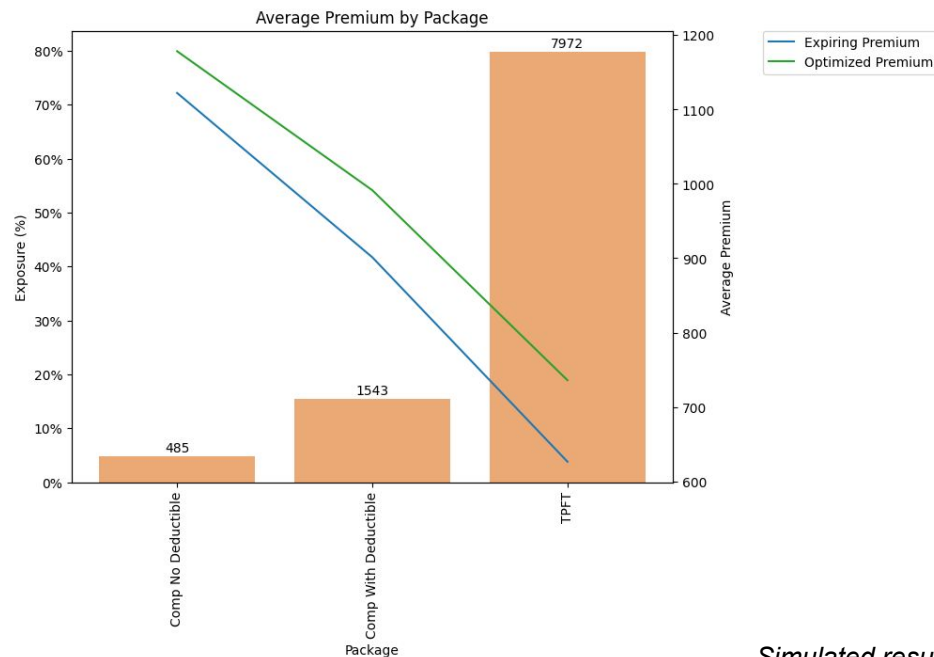
$$\frac{\sum_{\text{Policies}} \text{Retention Probability}}{n_{\text{Policies}}}$$

- **Expected Premium Retention:** $\sum_{\text{Policies}} \text{Optimized Premium} \times \text{Retention Probability}$

Final Result Evaluation

After the optimization is ready, we should see a **segmented view** to see which **segments** may now be getting larger increases, ensuring we are vaccinated against poodle bites.

In this example, the segments with the **highest average premium** are **not the ones with the highest increase** as, if they were, we would **lose all the agent policies**.



Insurance Price Optimization Summary and Challenges

- Margin Optimization problem is usually solved by **Lagrange Multipliers**.
- Very complex dimensionality (Big Data tool needed, with WTW Radar and Earnix being the most common).
- Lagrange Multipliers require a **Smooth & Differentiable Curve**, which Demand Models may not give, especially when they are Tree-Based Models.
- So, we need to make the **Estimated Demand Curves Smooth** in order to perform Price Optimization.
- Otherwise optimization will **get stuck** at local optima and turn the price distribution into a bimodal one instead of an approximately normal distribution.

After the optimization is ready, all business KPIs must be recomputed for the final Optimized Premium, namely **Profitability** (given by **Loss Ratio**) and **Growth** (given by **Policy & Premium Retention**), but also a segmented view to see which **segments** may now be getting larger increases, ensuring we are vaccinated against poodle bites.

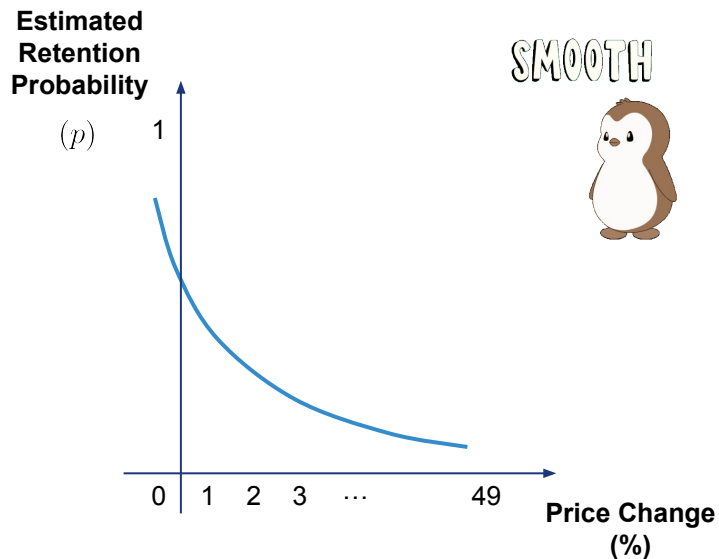


"Real-life" demonstration of a broker reaction to price increases.

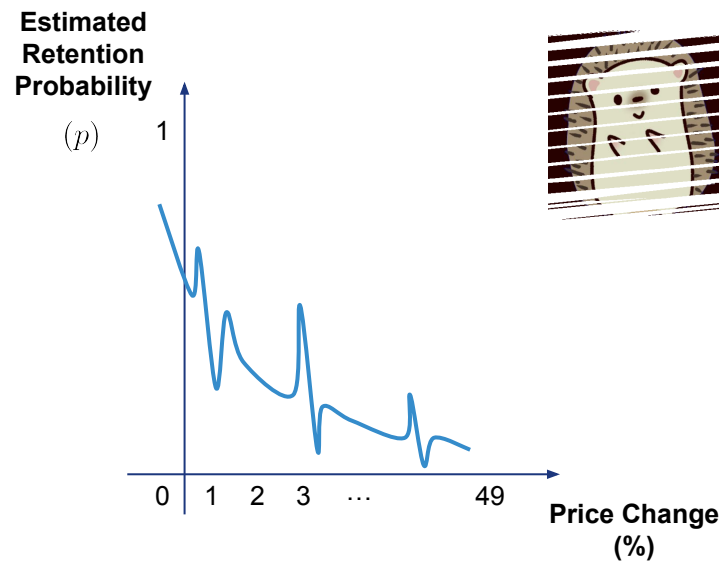
How to Smooth Models for Optimization?

Insurance Price Optimization - Spiky vs. Smooth Demand Curves

Expectation



Reality



The Need for a Smooth Operator: Turning Expectations into Reality

- This smoothing method allows for a **stable Price Optimization**, regardless of the base Machine Learning Model.
- It is **flexible** to adjust according to **business needs**.
- Provides predictions in **real-time**.
- It flexibly deals with different combinations of **multiple models**.
- This technique **alone** can contribute to a reduction of about **0.1pp** in Loss Ratio.
- Recommended improvement: to automate the **threshold selection** (based on hyperparameter tuning or fixing intended slopes).



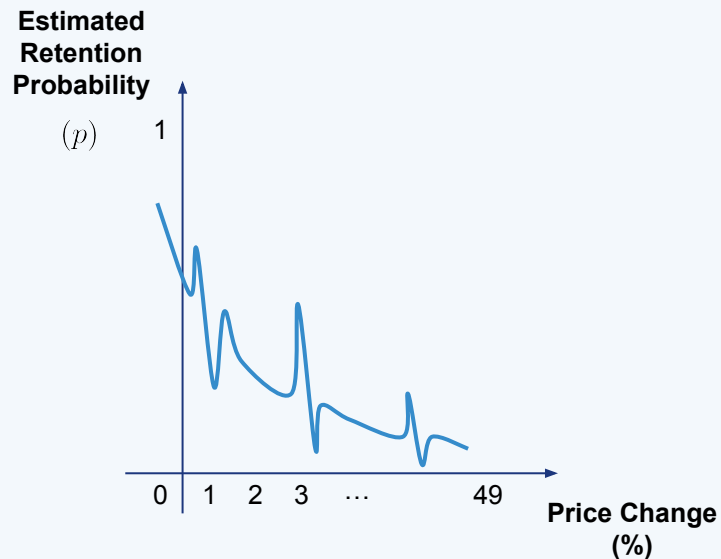
Smoothing Spiky Demand Models - Base Idea

1. For each policy, compute the retention probability at different price levels:

Create a matrix where every row corresponds to a policy and each column to the respective retention probability at a different price level within a specified range (e.g., 0%, 1%, ..., 49%).

	0%	1%	...	49%
Policy 1	0.95	0.89	...	0.55
Policy 2	0.75	0.68	...	0.37
...	...	...	...	...
Policy n	0.32	0.26	...	0.02

Retention Probability at Different Price Levels



Smoothing Spiky Demand Models - Base Idea

2. Map probabilities into real numbers using a logit transform:

For every probability in the matrix, apply the log-odds transformation.

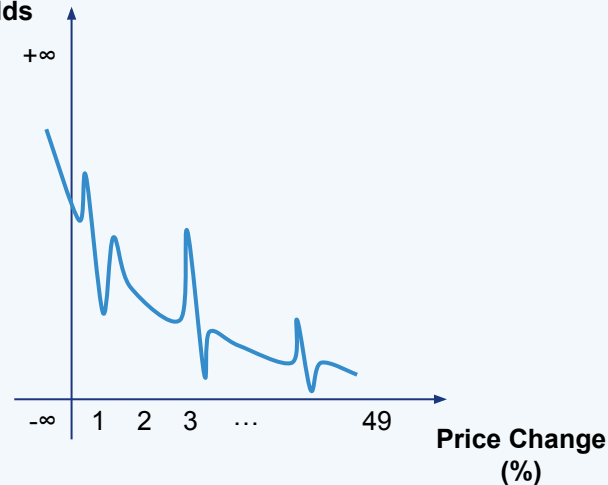
$$p_l = \log \left(\frac{p}{1-p} \right)$$

	0%	1%	...	49%
Policy 1	2.94	2.09	...	0.20
Policy 2	1.10	0.75	...	-0.53
⋮	⋮	⋮	⋮	⋮
Policy n	-0.75	-1.05	...	-3.89

Retention Log-Odds at Different Price Levels

Estimated
Retention
Log-Odds

(p_l) $+\infty$



(*) Shape of the plot changes, not shown for graphical simplicity.

Smoothing Spiky Demand Models - Base Idea

3. For each line/policy, fit a Linear Regression model, using exposure as weights:

For each line, fit a Linear Regression (e.g., third degree) to predict the log-odds of the retention probability using the offered price only.

$$p_l = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots + \epsilon$$

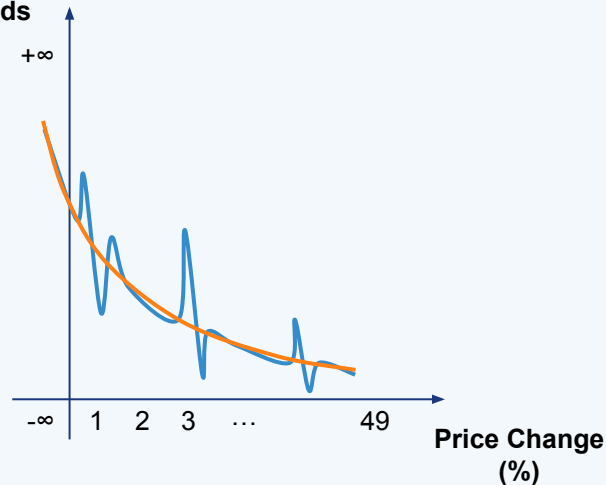
$$\hat{p}_l = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots$$

	0%	1%	...	49%	β_0	β_1	β_2	β_3
Policy 1	2.94	2.09	...	0.20	2.5	5.1	-100.7	164.6
Policy 2	1.10	0.75	...	-0.53	1.0	-6.1	5.4	1.8
...	...	...	...	...	...	...	...	...
Policy n	-0.75	-1.05	...	-3.89	-0.9	-4.4	-21.0	35.5

Smooth Retention Log-Odds at Different Price Levels

Estimated
Retention
Log-Odds

(p_l)



Smoothing Spiky Demand Models - Base Idea

4. Apply a sigmoid to reverse the logit and obtain smooth predictions:

Those will be the smooth demand predictions.

$$\hat{p} = \frac{1}{1 + e^{-\hat{p}_l}}$$

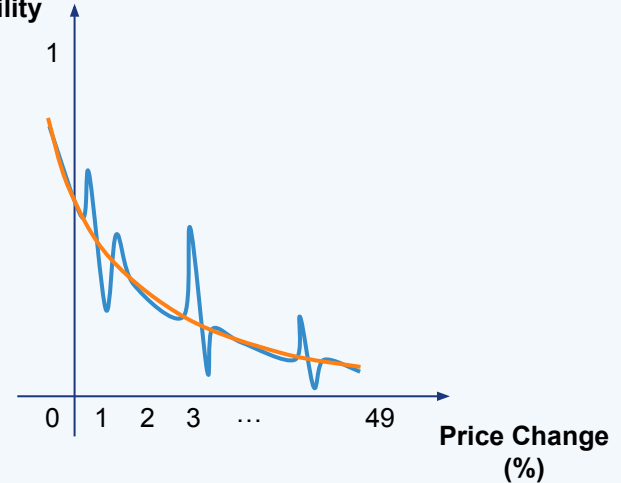


But fitting a Linear Regression per policy makes predictions too slow and not suited for real-time...

Smooth Retention Probability at Different Price Levels

Estimated
Retention
Probability

(p)



(*) Shape of the plot changes, not shown for graphical simplicity.

Smoothing Spiky Demand Models - Improvement for Real-Time Predictions

3.1. For each line/policy, fit a Linear Regression model, using exposure as weights:

For each line, fit a Linear Regression (e.g., third degree) to predict the log-odds of the retention probability using the offered price only.

$$\hat{p}_l = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \dots$$

3.2. Using the features from the original Spiky Demand Model, build similar ML Models (Beta Models) to predict the regression coefficients:

Using the same set of features X , train a model to predict β_0 , another for $\beta_1, \beta_2, \beta_3, \dots$ so that you can obtain the log-odds estimate in real time.

$$\hat{p}_l^* = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2 + \hat{\beta}_3 x^3 + \dots$$

Smooth Retention Log-Odds at Different Price Levels



Smoothing Spiky Demand Models - Improvement for Real-Time Predictions

4. Apply a sigmoid to reverse the logit and obtain smooth predictions:

Those will be the smooth demand predictions, using Beta Models.

$$\hat{p}^* = \frac{1}{1 + e^{-\hat{p}_i^*}}$$

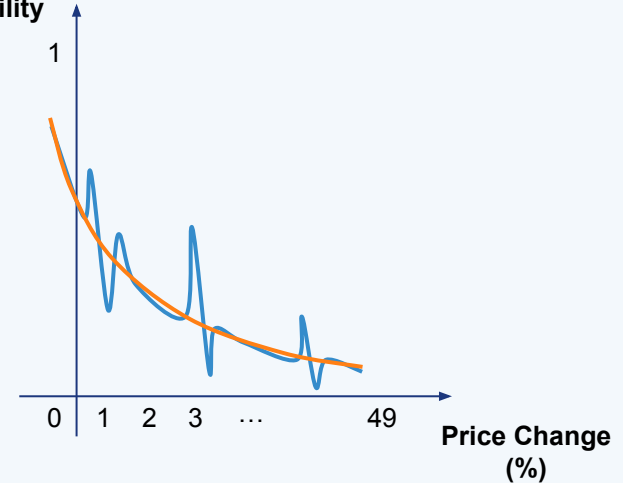


But a **Polynomial Curve** always reverses and may not be suited for a **Demand Curve**, which is naturally decreasing...

Smooth Retention Probability at Different Price Levels

Estimated Retention Probability

(p)



(*) Shape of the plot changes, not shown for graphical simplicity.

Smoothing Spiky Demand Models - Improvement for Business Adjustment

4. Define a Validity Price Domain and Apply Tangent Lines Outside:

An n -degree polynomial has n curves and countercurves, so the monotonicity will reverse, even though the demand curve must not.

Define a validity domain $[a, b]$ for which you will use the smooth predictions defined before.

Outside of the domain, use the tangent line to the smooth curve on the thresholds a and b .

$$\hat{p}_l^* = \begin{cases} \beta_0 + \beta_1 a + \beta_2 a^2 + \beta_3 a^3 + (\beta_1 + 2\beta_2 a + 3\beta_3 a^2)(x - a), & \text{if } x < a \\ \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3, & \text{if } a \leq x \leq b \\ \beta_0 + \beta_1 b + \beta_2 b^2 + \beta_3 b^3 + (\beta_1 + 2\beta_2 b + 3\beta_3 b^2)(x - b), & \text{if } x > b \end{cases}$$

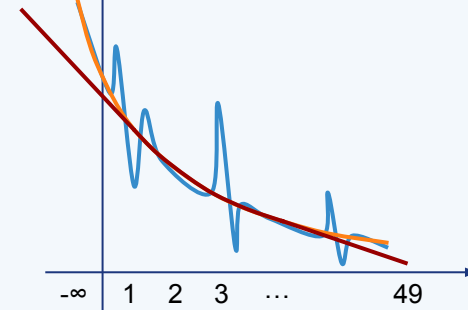
Smooth Retention Log-Odds at Different Price Levels

Estimated
Retention
Log-Odds

(p_l)

$+\infty$

Flatter curve on the left guarantees not too many discounts are sent



Steep increase on the right guarantees not too many very high prices are sent

Price Change
(%)

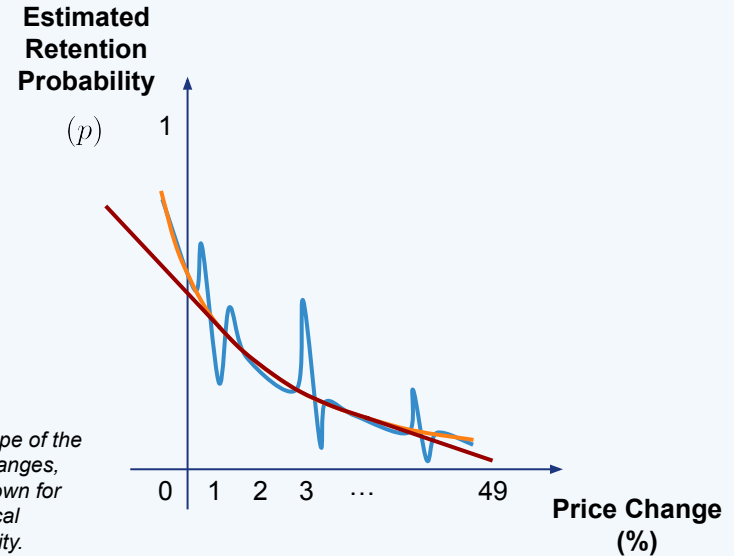
Smoothing Spiky Demand Models - Improvement for Business Adjustment

5. Apply a sigmoid to reverse the logit and obtain smooth predictions:

Those will be the smooth demand predictions, using Beta Models.

$$\hat{p}^* = \frac{1}{1 + e^{-\hat{p}_i^*}}$$

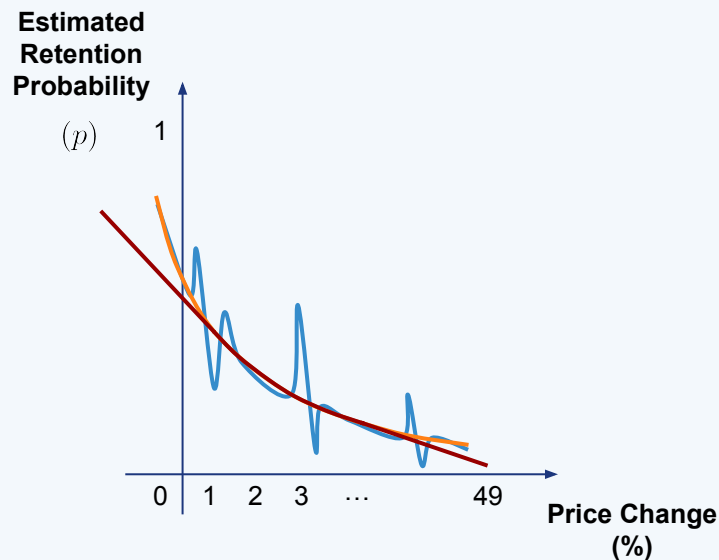
Smooth Retention Probability at Different Price Levels



Smoothing Spiky Demand Models - Summary

1. For each policy, compute the **retention probability at different prices levels**
2. Map probabilities into real numbers using a **logit transform**
3. For each **line/policy**, fit a **Linear Regression** model, using **exposure as weights**, and **train Beta Models** to predict the **regression coefficients**, using the same set of features
4. Define a **Validity Price Domain** and Apply **Tangent Lines Outside**
5. Apply a **sigmoid** to reverse the logit and obtain **smooth predictions**

Smooth Retention Probability at Different Price Levels



Optimization Result Analysis

For the same Average Price Increase (global constraint) and provided Caps & Collars (local constraints), check by how much:

- Margin
- Loss Ratio
- Policy Retention
- Premium Retention

vary. Ideally, they should all improve, but the scenario may change according to portfolio distribution and Caps & Collars. Moreover:

- **Price Distribution** is expected to be closer to a normally shaped distribution
- **Segment Analysis** must be performed to ensure small impactful segments are harming the business.

Expected KPIs	Before Optimization	After Optimization
Average Price Increase	10%	10%
Margin	€ 100 M	€ 102 M (+2%)
Loss Ratio	65.21%	64.96% (-0.25pp)
Policy Retention	78.12%	78.93% (+0.81pp)
Premium Retention	79.15%	79.99% (+0.84pp)

How Can We Transpose these Learnings to AUTO1?

Insurance Price Optimization Problem Redefined for AUTO1

Insurance:

- Determining the set of policy **Premiums** that **maximize**

$$\text{Margin} = \sum_{\text{Policies}} (\text{Premium} - \text{Costs}) \times \text{Retention Probability}$$

AUTO1:

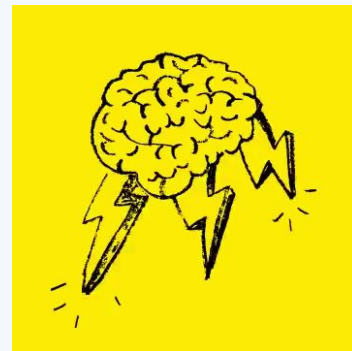
- Determining the set of **Sale Prices** that **maximize**

$$\text{Margin} = \sum_{\text{Cars}} (\text{Sales Price} - \underbrace{\text{Costs}}_{\text{incl. Onsite Price}}) \times \text{Conversion Probability}$$

with:

- Offered Onsite Price also having its own Conversion Probability;
- Inventory management needing to be included in the logic too.

Brainstorm!



Thank you!